

FRAUD ANALYTICS & INTELLIGENCE REPORT

ECHS FRAUD ANALYTICS

Module 14 — Pre-Authorization Deviation

PRE-AUTH ESTIMATE vs ACTUAL BILLED | LAST 5 YEARS | COST ESCALATION
& CONTROL BYPASS

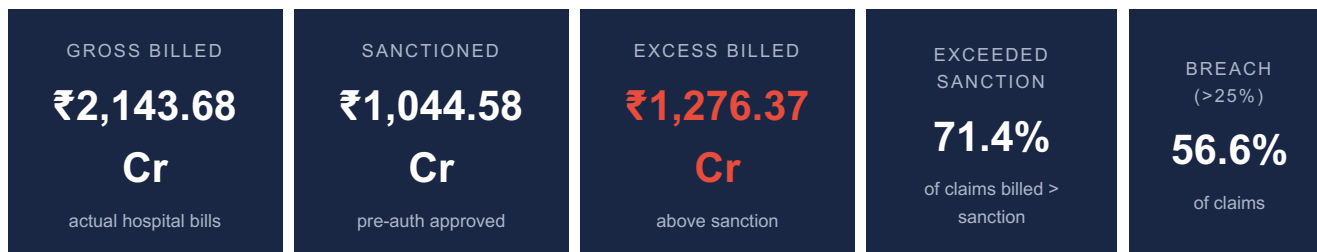
PRE-AUTH CLAIMS	GROSS BILLED	SANCTIONED	EXCESS BILLED	BILL / SANCTION
142,200	₹2,143.68 Cr	₹1,044.58 Cr	₹1,276.37 Cr	2.05×

IIT KANPUR — Data Analytics & Fraud Intelligence Division

05 June 2026 | Classification: RESTRICTED

Executive Summary

Module 14 examines the **pre-authorization** control: the gap between what a hospital declares (and ECHS sanctions) *before* treatment and what it actually **bills** afterwards. Over the last five years, **142,200** claims passed through the live unlisted-procedure pre-auth channel with both a sanction and a final bill. They were sanctioned for **₹1,044.58 Cr** but billed **₹2,143.68 Cr** — an overall **2.05×** ratio and **₹1,276.37 Cr** billed above sanction.



By composite risk band, **12** hospital logins (of 723 with ≥ 20 pre-auth claims) fall in the CRITICAL/HIGH tiers. The bill-to-sanction ratio has risen steadily from 1.82× to 2.07× across the period — a sustained escalation, not a one-off.

Patterns at a glance

#	Pattern	What it detects	Metric	Risk
1	Bill inflation above sanction	Hospitals billing far more than the pre-authorised amount	71.4% exceed	CRITICAL
2	Excess billed	Total actual bill beyond what was sanctioned	₹1,276.37 Cr	CRITICAL
3	Hospital systematic overbilling	Specific logins with large, consistent excess	723 logins	CRITICAL
4	Cost escalation	Bill/sanction ratio rising year on year	1.82→2.07×	HIGH
5	Rubber-stamp sanction	Sanction set equal to the hospital's own estimate (no scrutiny)	≈1.00× at stage 2	HIGH
6	Pre-auth coverage gap	High-value claims not routed through pre-auth (context)	5.9% covered	MEDIUM

Immediate recommended actions

- Audit the top-excess hospitals** (723 logins ≥ 20 claims) — reconcile a sample of bills against their pre-auth sanctions and CGHS package rates.
- Introduce partial sanction / CGHS benchmarking at Stage 2** — today the sanction simply equals the hospital's own estimate, so there is no financial gate.
- Investigate the steepest bill/sanction ratios** — they reached 2.07× on average and far higher per claim.
- Recover demonstrable overbilling** where the actual bill cannot be justified against the sanctioned, documented procedure.

HOW THE ANALYSIS WORKS

Parameters & Methodology

Data. The live pre-auth system (**unlisted_procedure**, Dec 2021 onward) joined to its claim (**claim_intimation** for the treating hospital via office master, **claim_submission** for the actual billed and UTI-approved amounts). Revised pre-auth rows are deduplicated to the latest per (claim, procedure), then summed per claim.

Five-stage pipeline. (1) hospital submits a pre-auth request with a cost estimate → (2) ECHS sanctions it (in practice, at the declared estimate) → (3) hospital treats and submits the actual bill → (4) UTI audits and approves → (5) net payment. Module 14 measures the gap between the **Stage-2 sanction** and the **Stage-3 actual bill**.

Table P.1 — Parameters & definitions

Parameter	Definition
Pre-auth sanction	SUM(UP_SANC_TOTAL) per claim — the amount ECHS authorised before treatment
Actual billed	CS_GR_CLAIM_AMT — the gross amount the hospital billed after treatment
Bill / Sanction ratio	actual billed ÷ pre-auth sanction (aggregate: total billed ÷ total sanction)
Excess billed	actual billed – sanction, summed over claims that exceeded sanction
Exceedance	share of claims with bill > sanction (ratio > 1.0)
Breach	bill > 1.25× sanction (the framework's '>25% above estimate')
Disallow %	(billed – UTI approved) / billed — the share the UTI audit later cut
Matched claim	a pre-auth claim with both a sanction (>0) and a final bill (>0)

Hospital risk score — component weights (0–100, no ML)

Bands: **CRITICAL** ≥ 70, **HIGH** 45–69, **MEDIUM** < 45. A plain weighted sum of normalised signals — fully reconstructable from the source rows.

Component	Weight
Excess Magnitude	35
Avg Ratio	30
Breach Rate	20
Volume	15

Limitations & Assumptions

- **A ratio above 1 is partly structural.** The pre-auth sanction covers only the *unlisted procedure*, while the final bill includes the whole admission (room, nursing, drugs, listed packages). So a bill above sanction is expected; what is anomalous is an **extreme and consistent** gap across many claims at one hospital.
- **Aggregate ratio, not mean-of-ratios.** We rank by absolute excess billed and report the aggregate bill/sanction ratio (total bill ÷ total sanction). Per-claim ratios can look far higher because some sanctions are tiny — we avoid that denominator inflation.
- **Coverage, not bypass.** Only 5.9% of >₹1L claims went through this pre-auth channel — because most are listed-package claims that do not require unlisted-procedure pre-auth. The 1,296,165 'no pre-auth' claims are a coverage observation, NOT a fraud count.
- **Identifiers, not verdicts.** Hospital/office names are identifiers; a high deduction or ratio is an investigative lead. Every case needs verification before any action.
- **Scope.** Primary analysis = last five years (Type-2 unlisted_procedure). The legacy pre_auth system (Type-1, 2012–2017) is summarised separately as historical context.

ALL CASES REQUIRE VERIFICATION BY QUALIFIED AUDITORS BEFORE ANY ACTION IS TAKEN.

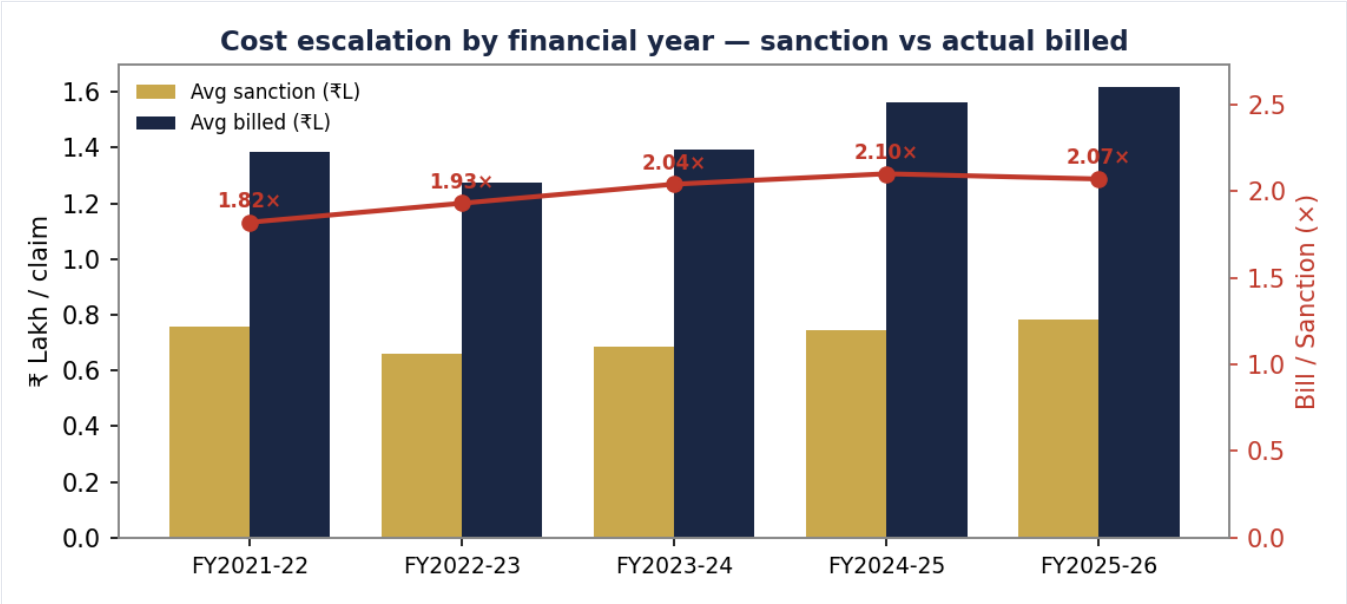
Q14C · YEAR-ON-YEAR

Cost Escalation

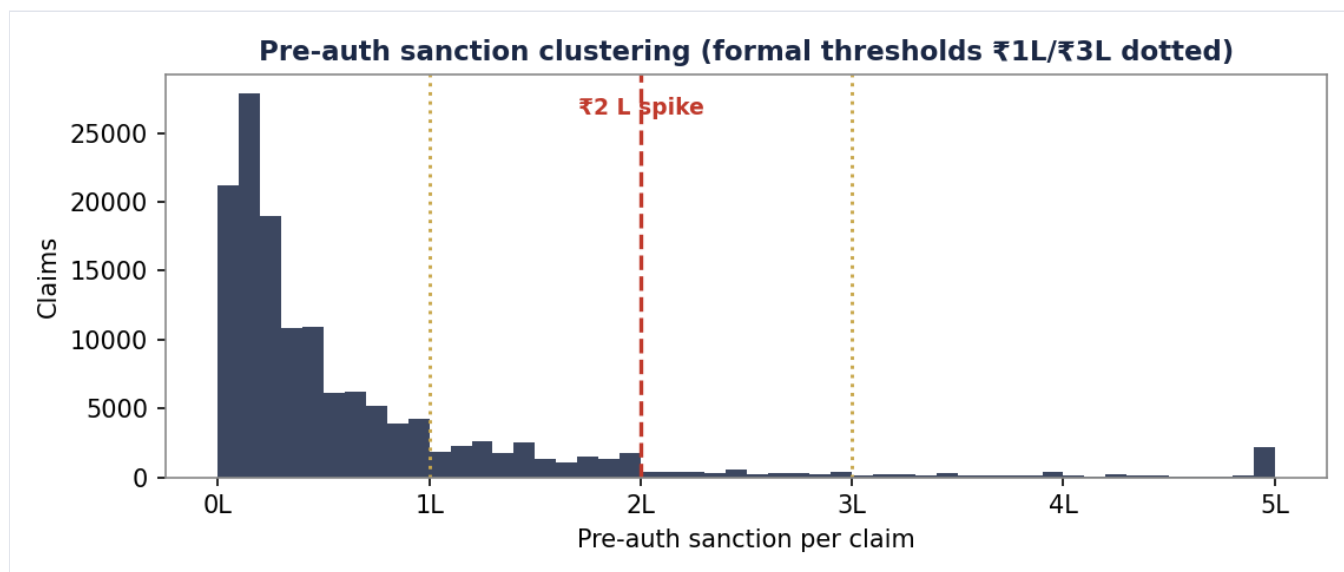
Bill-to-sanction gap by financial year

Table Q14c.1 — Average sanction vs actual billed, by financial year

FY	Claims	Avg Sanction	Avg Billed	Avg UTI Approved	Bill/Sanc
FY2021-22	1,251	₹75,921	₹1.38 L	₹1.26 L	1.82×
FY2022-23	18,393	₹66,009	₹1.28 L	₹1.16 L	1.93×
FY2023-24	31,044	₹68,368	₹1.39 L	₹1.18 L	2.04×
FY2024-25	38,932	₹74,625	₹1.56 L	₹1.26 L	2.1×
FY2025-26	52,580	₹78,147	₹1.62 L	₹90,186	2.07×



The average bill has pulled steadily away from the average sanction every year, while the sanction itself barely moves — consistent with hospitals treating the pre-auth as an entry token rather than a cost ceiling. Sanction amounts also cluster at round figures:



The spike at ₹2 lakh is notable: it is **not** a formal ECHS threshold (₹1 L and ₹3 L are), yet it draws more sanctions than either — suggesting estimates are deliberately set just under the level that would trigger higher scrutiny.

Q14A · PROVIDER-LEVEL

Hospital Deviation Scorecard — Top 20

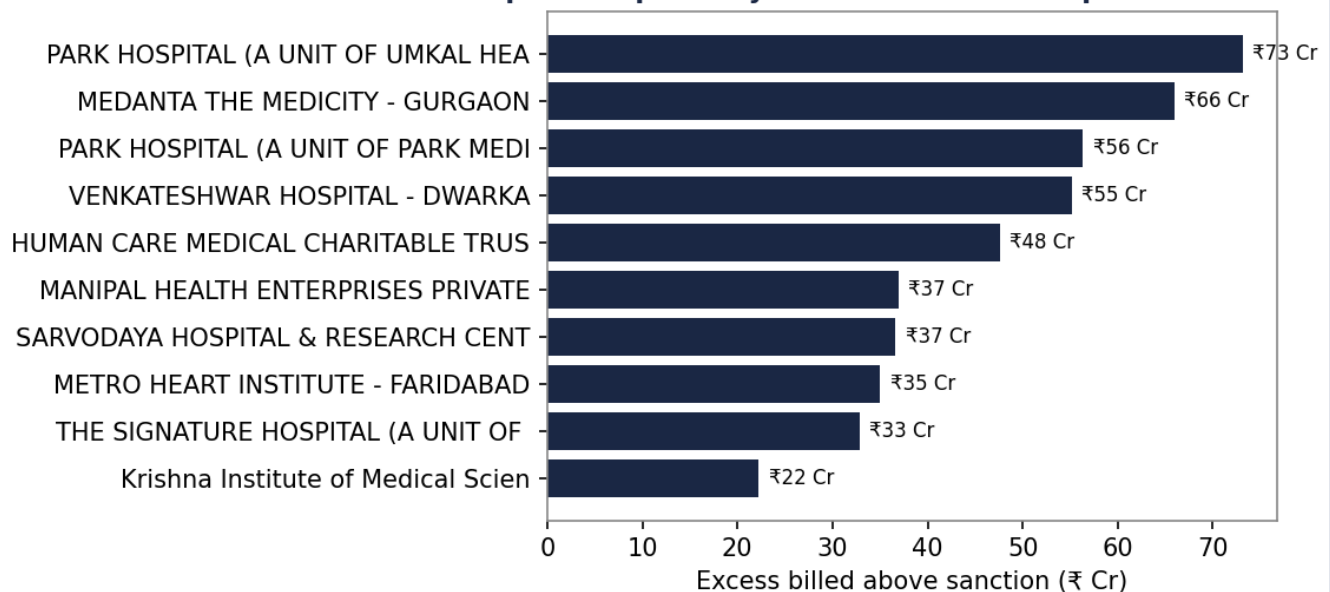
Pre-auth hospitals ranked by excess billed
above sanction

Table Q14a.1 — Top 20 hospitals by excess billed

Risk	Hospital	Claims	Billed	Excess	Avg B/S	Disallow	Breach
CRITICAL 76.4	PARK HOSPITAL (A UNIT OF UMKAL HEALTHCA...	1,748	₹88.69 Cr	₹73.11 Cr	5.56×	32.9%	1,639
HIGH 56.7	MEDANTA THE MEDICITY - GURGAON	5,961	₹122.44 Cr	₹65.99 Cr	1.6×	33.0%	2,622
HIGH 61.0	PARK HOSPITAL (A UNIT OF PARK MEDICENTR...	1,407	₹74.82 Cr	₹56.33 Cr	3.73×	40.6%	1,104
HIGH 60.2	VENKATESHWAR HOSPITAL - DWARKA	3,772	₹73.30 Cr	₹55.21 Cr	3.61×	31.8%	2,624
HIGH 55.6	HUMAN CARE MEDICAL CHARITABLE TRUST	3,564	₹65.42 Cr	₹47.61 Cr	3.24×	33.6%	2,476
HIGH 50.6	MANIPAL HEALTH ENTERPRISES PRIVATE LIMI...	2,077	₹52.52 Cr	₹36.97 Cr	2.83×	29.9%	1,642
HIGH 47.6	SARVODAYA HOSPITAL & RESEARCH CENTRE - ...	3,261	₹58.77 Cr	₹36.62 Cr	2.21×	18.7%	2,211
HIGH 48.7	METRO HEART INSTITUTE - FARIDABAD	1,434	₹52.35 Cr	₹35.00 Cr	2.81×	48.6%	1,112
HIGH 58.9	THE SIGNATURE HOSPITAL (A UNIT OF PARK ...	935	₹37.98 Cr	₹32.83 Cr	7.09×	25.9%	857
MEDIUM 39.2	Krishna Institute of Medical Sciences L...	2,520	₹32.35 Cr	₹22.23 Cr	2.59×	17.7%	1,465
MEDIUM 37.2	Amrita Institute of Medical Sciences An...	2,416	₹27.70 Cr	₹21.26 Cr	2.44×	25.9%	1,265
MEDIUM 38.8	Aditya Birla Health Services Ltd	1,316	₹29.73 Cr	₹19.68 Cr	2.3×	17.2%	937
MEDIUM 31.3	ARTEMIS MEDICARE SERVICES LIMITED - GUR...	2,412	₹53.25 Cr	₹17.73 Cr	1.25×	24.0%	1,065
MEDIUM 38.2	MAX SUPER SPECIALITY HOSPITAL - SHALIMA...	890	₹23.66 Cr	₹16.76 Cr	2.86×	21.4%	646
MEDIUM 39.9	FORTIS ESCORTS HOSPITAL (A UNIT OF FHL ...	1,377	₹26.41 Cr	₹15.65 Cr	2.43×	15.9%	1,162
MEDIUM 33.6	GRAPHIC ERA INSTITUTE OF MEDICAL SCIENC...	793	₹21.11 Cr	₹14.16 Cr	2.67×	47.9%	464
MEDIUM 29.7	MAX HEALTH CARE INSTITUTE LTD - DEHRADUN	2,474	₹28.85 Cr	₹13.50 Cr	1.67×	14.7%	1,032
MEDIUM 40.3	ACCORD SUPER SPECIALITY HOSPITAL	644	₹17.32 Cr	₹13.12 Cr	3.4×	33.7%	570

Risk	Hospital	Claims	Billed	Excess	Avg B/S	Disallow	Breach
MEDIUM 35.1	FORTIS HOSPITAL - SHALIMAR BAGH	615	₹17.29 Cr	₹12.32 Cr	2.91×	39.4%	431
HIGH 48.8	EMC Super Speciality Hospital Pvt. Ltd.	387	₹14.11 Cr	₹12.10 Cr	7.0×	24.0%	384

Top 10 hospitals by excess billed above pre-auth sanction



Hospital deep-dives

PARK HOSPITAL (A UNIT OF UMKAL HEALTHCARE PVT LTD) — 1,748 pre-auth claims, billed ₹88.69 Cr against ₹15.96 Cr sanctioned, i.e. **₹73.11 Cr excess** (aggregate 5.56× bill/sanction). The UTI audit later disallowed 32.9% of the billed amount, and 1,639 of its claims breached the 25% threshold — a consistent, hospital-wide pattern of billing well beyond the authorised estimate.

MEDANTA THE MEDICITY - GURGAON — 5,961 pre-auth claims, billed ₹122.44 Cr against ₹76.30 Cr sanctioned, i.e. **₹65.99 Cr excess** (aggregate 1.6× bill/sanction). The UTI audit later disallowed 33.0% of the billed amount, and 2,622 of its claims breached the 25% threshold — a consistent, hospital-wide pattern of billing well beyond the authorised estimate.

PARK HOSPITAL (A UNIT OF PARK MEDICENTRES & INSTITUTIONS PVT LTD) - GURGAON — 1,407 pre-auth claims, billed ₹74.82 Cr against ₹20.04 Cr sanctioned, i.e. **₹56.33 Cr excess** (aggregate 3.73× bill/sanction). The UTI audit later disallowed 40.6% of the billed amount, and 1,104 of its claims breached the 25% threshold — a consistent, hospital-wide pattern of billing well beyond the authorised estimate.

Q14A · CLAIM-LEVEL

Top High-Deviation Claims

Individual claims with the largest gap between sanction and actual bill

Table Q14a.2 — Top 20 claims by excess billed above sanction

Beneficiary	Card	Hospital	Sanctioned	Billed	B/S
HEM JOSHI	000008854010	MAYO MEDICAL CENTRE PVT LTD	₹5.61 L	₹99.48 L	17.74×
ROHTASH KUMAR	DL1000002724960	METRO HEART INSTITUTE - FAR...	₹1.02 L	₹73.32 L	71.53×
PAWAN KUMAR	000009088126	AMRITA HOSPITAL	₹1.81 L	₹65.48 L	36.26×
SURESH CHANDRA UPADHY...	DH000003043976	GRAPHIC ERA INSTITUTE OF ME...	₹1.95 L	₹55.86 L	28.65×
BRAJ NARAYAN SINGH	JB000004358846	MEDANTA THE MEDICITY - GURG...	₹0.50 L	₹52.63 L	104.94×
VEER SINGH GANGWAR	BR000006499925	MEDANTA THE MEDICITY - GURG...	₹2.62 L	₹50.48 L	19.25×
SUMER SINGH	000004738364	FORTIS HOSPITAL LTD - JAIPUR	₹0.22 L	₹45.87 L	210.01×
NARAIN SINGH	JA000005774061	METRO HEART INSTITUTE - FAR...	₹0.30 L	₹45.78 L	152.61×
COSMOS SYLVESTER S	BA000007557145	ST. JOHNS MEDICAL COLLEGE A...	₹0.15 L	₹45.13 L	300.88×
MAHABIR SINGH YADAV	JA000002247343	NORTH EAST HEALTHCARE PVT L...	₹2.75 L	₹46.67 L	16.97×
TEJ RAM	DL1000002662028	METRO HEART INSTITUTE - FAR...	₹3.82 L	₹47.32 L	12.38×
DAMBAR SINGH	BR000002245802	MEDANTA THE MEDICITY - GURG...	₹0.13 L	₹43.20 L	333.11×
SATISH KUMAR	DL1000002421736	MEDANTA THE MEDICITY - GURG...	₹0.56 L	₹41.40 L	73.43×
KIRAN PAL SINGH	DL1000008254523	VENKATESHWAR HOSPITAL - DWA...	₹1.34 L	₹41.75 L	31.11×
SEWA SINGH GILL	HI000003702136	MEDANTA THE MEDICITY - GURG...	₹3.42 L	₹42.89 L	12.54×
GAJENDER SINGH TOMAR	000004383368	METRO HEART INSTITUTE - FAR...	₹2.13 L	₹41.35 L	19.37×
AKSHYA KUMAR DAS	000002871145	QRG MEDICARE LIMITED	₹0.23 L	₹39.14 L	167.77×
SHIV DUTT	000006410216	SARVODAYA HOSPITAL & RESEAR...	₹9.69 L	₹48.38 L	4.99×
M MANGI SINGHA	000003895764	AMRITA HOSPITAL	₹1.34 L	₹39.79 L	29.59×

Beneficiary	Card	Hospital	Sanctioned	Billed	B/S
NANDI RAM	DL100000577629 9	VENKATESHWAR HOSPITAL - DWA...	₹1.63 L	₹40.04 L	24.64×

Note: a large gap can reflect a small unlisted-procedure sanction against a large whole-admission bill; these are leads for line-item verification, not confirmed overbilling.

Q14B · CONTEXT

Pre-Auth Coverage & Control Bypass

How much of the high-value claim flow actually passes through pre-auth

Of 1,378,161 high-value claims (>₹1 lakh) in the last five years (₹31,702.55 Cr gross), only 5.9% went through the unlisted-procedure pre-auth channel. The remaining 1,296,165 (₹29,259.12 Cr) carry no pre-auth record.

Read this as coverage, not fraud. ECHS pre-auth (unlisted procedure) is required only for specific non-package procedures; the vast majority of high-value claims are listed-package admissions that legitimately need no pre-auth. The figure simply shows how narrow the pre-auth net is — and therefore how few high-value claims are subject to any pre-treatment cost check.

Table Q14b.1 — Largest high-value claims with no pre-auth record (sample)

Beneficiary	Card	Hospital	Billed	UTI Approved
SURENDRA SINGH	PT000006563832	Ara Bhojapur	₹86.88 L	₹0
HARISH KUMAR SHARMA	MU000004121457	Mumbai - Colaba	₹74.14 L	₹0
AMRIT LAL CHADHA	DL1000004517957	Delhi Cantt (BHDC)	₹67.16 L	₹0
RAMESH NAGPAL	DL2000006699939	Lodhi Road	₹66.53 L	₹0
NARESH KUMAR BUDHWAR	HI000006658647	Rohtak	₹63.67 L	₹25.00 L
ARUN KUMAR SHOREY	DL1000004878364	Gurgaon (Dhundaheera)	₹63.64 L	₹0
NIRANJAN DAS	BU000007183756	Bhubaneswar (RC Bhubneshwar)	₹62.24 L	₹0
APURBA MAITY	MU000002562990	Nerul	₹58.11 L	₹0
OM SINGH	000004552995	HEALTHCARE GLOBAL ENTERPRIS...	₹58.10 L	₹0
RAJARAM RAJA SANJAY	CH000002350873	Chennai	₹56.21 L	₹13.06 L
M S GANGA RAO	BA000004372854	Bangalore (Deactivated)	₹55.68 L	₹0
PREM SHANKAR TEWARI	MU000007835581	Nerul	₹55.30 L	₹0

2012–2017

Legacy Pre-Auth System (Type-1) — Historical Note

The discontinued pre_auth table, for context

Before the current system, ECHS ran a separate **pre_auth** table (2012–2017, now discontinued). It held just **2,949** records, of which **97.3%** were marked rejected (PA_APPROVED = N) — yet historical analysis showed rejected pre-auths were still settled and paid at almost the same rate as approved ones. In other words, the legacy approval gate had little bearing on payment. It is included here only as historical context; the last-five-year analysis above is based entirely on the live unlisted-procedure system.

Table T1.1 — Legacy pre_auth decisions (2012–2017)

Decision	Claims	Est. cost (₹L)
N — Rejected	2,869	₹341.27 L
Y — Approved	65	₹4.13 L
X — Cancelled	15	₹0.16 L

Conclusions

The pre-authorization control, as currently operated, does not constrain cost. Across 142,200 pre-auth claims in the last five years, hospitals billed **₹2,143.68 Cr** against **₹1,044.58 Cr** sanctioned — **₹1,276.37 Cr** in excess, with **71.4%** of claims billed above their sanction.

The root cause is structural: ECHS sanctions at the hospital's own declared estimate, with no partial sanction, CGHS benchmarking or cost negotiation at Stage 2. The pre-auth therefore functions as an entry token, not a ceiling — and the gap has widened every year. The concentration of excess at a small set of hospitals, and the deliberate clustering of estimates just under scrutiny thresholds, are the clearest leads for audit.

The fix is a financial gate at Stage 2 (partial sanction against CGHS/package rates), backed by post-hoc audit of the top-excess hospitals. As always, these are prioritised leads — each must be confirmed against itemised bills and clinical records before recovery or enforcement.

Consolidated Risk Register & Recommendations

Table R.1 — Risk register

Signal	Scale	Exposure	Risk
Excess billed above sanction	71.4% of claims	₹1,276.37 Cr	CRITICAL
Hospital systematic overbilling	723 logins	see scorecard	CRITICAL
Cost escalation (year on year)	1.82→2.07×	widening gap	HIGH
Rubber-stamp sanction (Stage 2)	≈1.00× sanction:estimate	no financial gate	HIGH
Pre-auth coverage gap	5.9% of >₹1L covered	₹29,259.12 Cr outside	MEDIUM
Legacy approval gate (Type-1)	97.3% rejected yet paid	historical	MEDIUM

Table R.2 — Strategic recommendations (priority · SLA · target)

Priority	SLA	Recommendation	Action
CRITICAL	30 days	Audit the top-excess hospital	PARK HOSPITAL (A UNIT OF UMKAL HEALTHCARE PVT LTD) — billed ₹88.69 Cr vs sanctioned, ₹73.11 Cr excess (5.56×). Reconcile a 25-claim line-item sample against sanctioned procedures and CGHS rates.
CRITICAL	60 days	Install a financial gate at Stage 2	Replace rubber-stamp sanctioning (sanction = estimate) with partial sanction against CGHS/package benchmarks and a documented cost-negotiation step.
HIGH	60 days	Profile the top-20 deviation hospitals	All 723 logins with ≥20 pre-auth claims and high excess — empanelment review + billing audit; suspend the worst pending audit.
HIGH	90 days	Investigate estimate clustering at ₹2 L	The ₹2-lakh spike is not a formal threshold; review why estimates bunch there and whether it is used to stay under scrutiny limits.
MEDIUM	Ongoing	Widen pre-auth coverage of high-value claims	Only 5.9% of >₹1L claims pass any pre-treatment check; extend pre-auth to more high-cost admissions.
MEDIUM	Ongoing	Operationalise the deviation score	Recompute the hospital deviation score each cycle and route CRITICAL-band hospitals to audit before settlement.

This report is produced by an automated screening system. Every flagged case is an investigative lead, not a confirmed finding, and must be reviewed by a qualified auditor before any action is taken.

Prepared by IIT Kanpur — Data Analytics & Fraud Intelligence Division | 05 June 2026 | RESTRICTED